

TaxoBell: Gaussian Box Embeddings for Self-Supervised Taxonomy Expansion

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Abstract

Taxonomies form the backbone of structured knowledge representation across diverse domains, enabling applications such as e-commerce and semantic search. Yet, manual taxonomy expansion is labor-intensive and slow. Existing methods rely on point-based vector embeddings, which model symmetric similarity and thus struggle with the asymmetric relationships that are fundamental to taxonomies. Box embeddings offer a promising alternative by enabling containment and disjointness, but they face key issues: (i) unstable gradients at the intersection boundaries, (ii) no notion of semantic uncertainty, and (iii) limited capacity to represent polysemy or ambiguity. We address these shortcomings with TaxoBell, a Gaussian box embedding framework that translates between box geometries and multivariate Gaussian distributions, where means encode semantic location and covariances encode uncertainty. Energy-based optimization yields stable optimization, robust modeling of ambiguous concepts, and interpretable hierarchical reasoning. Extensive experiments on five benchmark datasets demonstrate that TaxoBell significantly outperforms eight state-of-the-art taxonomy expansion baselines by 19% in MRR and around 25% in Recall@k. We further demonstrate the advantages and pitfalls of TaxoBell with error analysis and ablation studies.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning.

Keywords

Taxonomy, Box Embedding, Gaussian Representation, Energy-Based Learning

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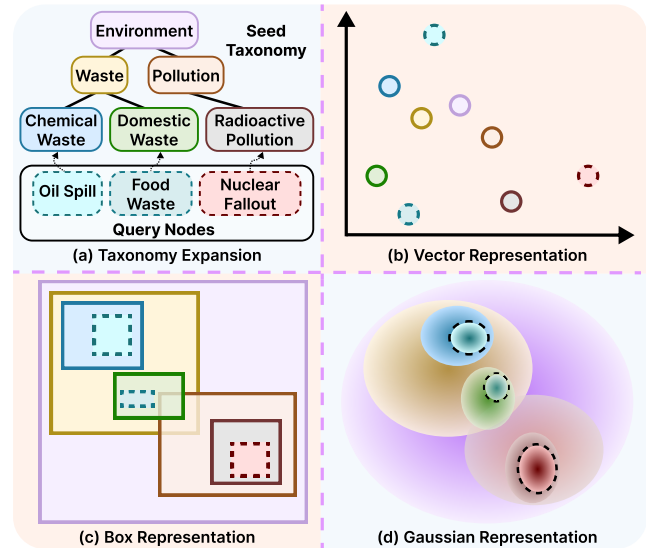


Figure 1: Overview of taxonomy expansion and TaxoBell.

1 Introduction

Taxonomies are domain-centric hierarchical structures that encode hypernymy (“is-a”) relations among concepts and entities, underpinning various applications. E-commerce platforms such as Amazon and Alibaba organize products to streamline navigation and retrieval [16, 23, 26, 46], while Pinterest curates taxonomies of *home decor* and *fashion styles* to improve visual search [24].

Despite their utility, most real-world taxonomies are still built and maintained by domain experts. This manual process is slow, expensive, and increasingly complex to sustain as data grows and new concepts appear every day. In practice, coverage lags behind what applications need, which can harm search quality, recommendation accuracy, and user experience. To keep pace, taxonomies must evolve continuously and at scale. This motivates automated *taxonomy expansion* in which, given an existing “seed” taxonomy, the goal is to add new entities, also called “query nodes”, by placing each one under the most appropriate parent node. Automated expansion reduces human effort, shortens update cycles, and helps prevent taxonomies from becoming stale in fast-changing domains. An instance of taxonomy expansion is shown in Fig. 1(a), where query nodes “*Oil Spill*”, “*Food Waste*” and “*Nuclear Fallout*” are placed under the most appropriate parents, which are “*Chemical*

Waste, *Domestic Waste* and *Radioactive Pollution*. For consistency, we refer to the child as the query and the parent as the anchor in accordance with prior work [29, 33, 39].

Summary of existing work on taxonomy expansion. In taxonomy expansion, methods typically fall into two streams, namely *semantic* and *structural* methods. Early semantic approaches infer hypernymy from lexical patterns such as Hearst [11, 30, 35] or distributional embeddings [6, 28]. More recent works adopt self-supervision, harvesting (parent, child) pairs from a seed taxonomy to train hypernymy classifiers; some methods, such as HiExpan [34] and STEAM [44], rely on corpus-derived features, while others, such as leveraging external textual descriptions of surface forms. However, they are limited by insufficient training data and fall short of fully exploiting the taxonomy’s inherent structural information. To incorporate structure, contemporary methods such as LOReX [27], TEMP [22] and TaxoEnrich [14] add signals from full taxonomy paths, STEAM [44] utilizes mini-paths, and TaxoExpan [33] and HEF [39] use local ego-graphs. Despite these advances, a core limitation of these methods is that they embed entities in Euclidean space as vectors, which are agnostic to asymmetric relationships and hierarchy possessed by the taxonomic geometry as shown in Fig. 1(b). To better align representation with asymmetric structure, a parallel line of work models entities with *box embeddings* [15], where each concept is an axis-aligned hyperrectangle and hypernym pairs are expressed through *containment*, meaning a child’s region lies entirely within its parent’s as shown in Fig. 1(c).

Limitations of existing work. While box embeddings capture asymmetric “is-a” structure via directional containment, which point (vector) embeddings cannot do, current formulations such as BoxTaxo [15] and TaxBox [43] remain limited for several practical reasons. First, their geometric training objectives defined over the intersection are typically piecewise and often suffer from vanishing or unstable gradients at disjoint boundaries, yielding weak or noisy learning signals for both centers and offsets. Secondly, boxes provide no principled way to represent *interpretable uncertainty* as their boundaries are hard margins, so ambiguity, polysemy, and noisy evidence cannot be expressed as calibrated confidence over space. Therefore, the core challenge is to achieve stable, asymmetric hierarchy modeling and principled uncertainty representation within a unified framework.

Our contributions. We model taxonomy entities as *Gaussian boxes*, which are axis-aligned hyperrectangles equipped with a multivariate Gaussian density that captures semantic location using *mean* (μ) and concept generality using *covariance* (Σ). Unlike mapping a point vector to a Gaussian distribution, which lacks hierarchical inductive bias and can fake inclusion by inflating or rotating covariance, Gaussian boxes occupy a geometric region and, unlike a hard box, assign calibrated probability mass within the region as shown in Fig. 1(d). This enables richer asymmetric relations such as *probabilistic containment*, *probabilistic disjointness*, and *partial overlap*. Moreover, the desired level of uncertainty is naturally controllable via confidence intervals, where the 1σ , 2σ , and 3σ enclose roughly 68%, 95%, and 99.7% of the probability mass, respectively, letting us set containment and overlap criteria at a chosen level and derive axis-wise box extents from Σ . We optimize the Gaussian boxes with energy-based objectives rather than probabilistic, geometric, or dot-product objectives on means, which fail

to incorporate covariances [38]. In contrast, energy functions score pairs of Gaussians by taking inner products between the distributions themselves (including covariance) and providing smooth, closed-form signals for Gaussians. Therefore, we define two energy functions: one for symmetric overlap to model semantic coherence and one for asymmetric overlap to learn hierarchical containment.

In this paper, we introduce TaxoBell¹, which translates between axis-aligned boxes and multivariate Gaussian parameterization and optimizes them with self-supervised, energy-based learning to capture semantic, asymmetric “is-a” relations, preserving calibrated uncertainty through the covariance, enabling robust placement of unseen entities for taxonomy expansion.

Specifically, we make the following contributions.

First, we use self-supervision to derive training data directly from the seed taxonomy, without extra annotations. Each (child, parent) edge forms a positive instance, while negatives are sampled from non-ancestors in the child’s neighborhood. This forces the model to separate the true parent from overly general ancestors and local confounders, yielding strong supervision from taxonomy structure at minimal cost.

Secondly, we project (child, parent) pairs to Gaussian boxes and jointly train them with two complementary energy functions, inspired by [38]. A symmetric overlap Bhattacharyya distance term encourages high probabilistic overlap for true pairs, while an asymmetric KL term enforces directional containment by pulling the child distribution inside its parent. To avoid degenerate solutions, we add volume regularization that keeps covariances well-conditioned and prevents variance collapse. At inference time, candidate parents are ranked by these energy functions, and the chosen Gaussian can be translated to a box by fixing a confidence level to control retained uncertainty.

Thirdly, we conduct an extensive set of experiments on five real-world taxonomy benchmarks against eight state-of-the-art baselines. Results demonstrate that TaxoBell consistently outperforms vector and geometric representations, such as box embeddings, improving Mean Rank (MR) by **43%**, Mean Reciprocal Rank (MRR) by **19%**, Recall@k by **25%**, and Hit@k by **21%**. We further present ablations on the projection design and energy-based optimization, along with case studies showing Gaussian-box representations and efficient attachment to the seed taxonomy.

2 Related Work

Taxonomy Expansion. Prior work expands taxonomies in two main ways. Corpus-based methods extract hypernyms from textual distance or contextual similarity [12, 19, 34], but require suitable corpora and often treat expansion as edge classification with limited global hierarchy awareness. Seed-taxonomy methods instead learn structure from existing paths [14, 22, 33, 39, 41, 44]: ego-network approaches focus on local context [33, 39], while path/walk models capture broader organization [22, 41]. Yet most represent nodes as points, which favors symmetric similarity and weakly encodes asymmetric hypernymy. We use Gaussian box embeddings to directly model directional containment, better aligning with taxonomy structure.

¹The name TaxoBell highlights the bell-shaped curve of the Gaussian distribution. Code: <https://github.com/sahilmishra0012/TaxoBell>.

Box Representation. Box embeddings naturally encode asymmetric hierarchy better than point vectors [20, 31, 37]. Gumbel “soft-edge” boxes ease optimization but blur crisp, human-interpretable margins [8]. Prior work shows geometry can improve interpretability [7, 32], and Jiang et al. [15] blends these ideas for taxonomy modeling. However, intersection-based box losses remain piecewise and fragile near boundaries, with uncertainty only implicit. We maintain the box’s inductive bias but introduce an explicit Gaussian density, yielding smooth, closed-form containment energies and calibrated per-dimension uncertainty while preserving the asymmetry crucial for taxonomy expansion.

Gaussian Representation. Gaussian embeddings model each concept as a distribution: the mean encodes its position, while the covariance captures uncertainty, generality, or polysemy. Early work showed that closed-form objectives such as expected likelihood and KL divergence let Gaussians represent confidence and asymmetric entailment [38]. Later extensions introduced multi-sense/mixture models [1, 2] and applied Gaussians to knowledge graphs (KG2E) to handle confidence and one-to-many relations [10]. Probability product kernels connect Gaussian overlap to positive-definite similarity [13], while KL naturally enforces directional containment. Energy-based training is smooth, jointly shaping means and covariances and avoiding brittle, piecewise losses [18]. Building on these ideas, we map axis-aligned boxes to Gaussians and optimize paired energies for semantic similarity and asymmetric containment with calibrated uncertainty.

3 Preliminaries

3.1 Taxonomy Expansion

Definition 3.1. Taxonomy. A taxonomy $\mathcal{T} = (\mathcal{N}, \mathcal{E})$ is defined as a directed acyclic graph where each node $n \in \mathcal{N}$ represents a concept with a surface name (a word or a phrase) while $\langle n_p, n_c \rangle \in \mathcal{E}$ represents an edge in the graph which represents a hypernymy relationship from parent n_p to child n_c .

Definition 3.2. Taxonomy Expansion. The taxonomy expansion task is to add a set of terms $\mathcal{C}$ to the given *seed taxonomy* $\mathcal{T}^0 = (\mathcal{N}^0, \mathcal{E}^0)$ such that it returns an expanded taxonomy $\mathcal{T} = (\mathcal{N}, \mathcal{E}) = (\mathcal{N}^0 \cup \mathcal{C}, \mathcal{E}^0 \cup \mathcal{R})$ where $\mathcal{R}$ represents newly added relations between seed nodes in $\mathcal{N}^0$ and new nodes in $\mathcal{C}$. Specifically, for each query node $n_q \in \mathcal{C}$, the model treats every seed node $a \in \mathcal{N}^0$ as an anchor, computes a matching score $f(a, n_q)$, and selects the best parent $n_p = \arg \max_{a \in \mathcal{N}^0} f(a, n_q)$. The edge $\langle n_p, n_q \rangle$ is then added to $\mathcal{R}$, expanding the seed taxonomy $\mathcal{T}^0$ to $\mathcal{T}$.

3.2 Box Representation

A box is parameterized by a pair of d -dimensional vectors $b = (c, o)$, where the *center* $c \in \mathbb{R}^d$ and the *offsets* $o \in \mathbb{R}^{d^+}$ define the axis-aligned hyperrectangle $\prod_{i=1}^d [c_i - o_i, c_i + o_i]$ [15, 32]. The positivity constraint $o_i > 0$ in every dimension ensures a non-degenerate box. Its volume is $\text{Vol}(b) = \prod_{i=1}^d (r_i - l_i)$, where $r_i = c_i + o_i$ and $l_i = c_i - o_i$. Following Jiang et al. [15], we define three pairwise relationships between the boxes: (i) **Enclosure**: $b_x \cap b_y = b_x$, where b_y encloses b_x ; (ii) **Intersection**: $b_x \cap b_y = b_z \neq \emptyset$, where b_z is the intersection box of b_x and b_y and has offset greater than 0; (iii) **Disjointness**: $b_x \cap b_y = b_z = \emptyset$, where b_z is an imaginary

intersection box of b_x and b_y and has offset less than or equal to 0. The intersection of the boxes b_x and b_y also leads to the computation of the conditional probability between these two boxes as, $P(b_y | b_x) = P(b_x, b_y) / P(b_x) = \text{Vol}(b_x \cap b_y) / \text{Vol}(b_x)$.

3.3 Multivariate Gaussian Preliminaries

Gaussian distribution. A multivariate Gaussian distribution $g = (\mu, \Sigma)$ can be described by its mean vector $\mu \in \mathbb{R}^d$ and symmetric positive definite covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$. A random variable $X \in \mathbb{R}^d$ is *Gaussian* ($X \sim \mathcal{N}(\mu, \Sigma)$) if its density $\mathcal{N}(x | \mu, \Sigma)$ is,

$$p(x) = \frac{1}{(2\pi)^{d/2} \sqrt{\det \Sigma}} \exp \left[-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right], \quad (1)$$

where the mean μ specifies the *location* of the variable in embedding space, and the covariance Σ controls its *uncertainty* along each direction, analogous to an offset vector. Intuitively, for a Gaussian, about 68%, 95%, and 99.7% of the mass lies within 1σ , 2σ , and 3σ of the mean (the “68–95–99.7 rule”). For computational stability we use axis-aligned Gaussians with $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_d^2) \succ 0$, so inverses and log-determinants are elementwise and $O(d)$.

Energy functions. An *energy function* [18, 38] $E_\theta(x, y)$ assigns a scalar score to an input–output pair (x, y) , parameterized by θ . The goal of energy-based learning is to tune θ so *positive* pairs have lower energy than *negatives* under a contrastive loss $\mathcal{L}$. For Gaussian embeddings, interactions between two distributions $P = \mathcal{N}(\mu_1, \Sigma_1)$ and $Q = \mathcal{N}(\mu_2, \Sigma_2)$ are generally captured by two kinds of complementary energy functions,

- A *symmetric similarity function*, which computes the dot product between the means of P and Q but does not incorporate the covariances and therefore discards probabilistic information. A more principled choice for a symmetric energy function is the inner product of the distributions themselves as $k(P, Q) = \langle P, Q \rangle = \int_{\mathbb{R}^d} P(x) Q(x) dx$ which is the $\alpha=1$ case of the *probability product kernel* (PPK) family [13], $k_\alpha(P, Q) = \int_{\mathbb{R}^d} P(x)^\alpha Q(x)^\alpha dx$ for $\alpha > 0$. Setting $\alpha = \frac{1}{2}$ yields the *Bhattacharyya coefficient*,

$$\text{BC}(P, Q) = k_{1/2}(P, Q) = \int_{\mathbb{R}^d} \sqrt{P(x) Q(x)} dx \in (0, 1], \quad (2)$$

and its Bhattacharyya distance $D_B(P, Q) = -\log \text{BC}(P, Q)$. Both $k(P, Q)$ and $\text{BC}(P, Q)$ increase with greater symmetric overlap, i.e., when the means are closer and the covariances are more compatible. BC has the closed form with,

$$D_B(P, Q) = \frac{1}{8} (\mu_1 - \mu_2)^\top \Sigma_m^{-1} (\mu_1 - \mu_2) + \frac{1}{2} \log \frac{\det \Sigma_m}{\sqrt{\det \Sigma_1 \det \Sigma_2}}, \quad (3)$$

where $\Sigma_m = \frac{1}{2} (\Sigma_1 + \Sigma_2)$. Hence, for Gaussian densities P and Q , we have $\text{BC}(P, Q) = \exp(-D_B(P, Q))$, providing a smooth, probabilistically grounded symmetric similarity.

- An *asymmetric energy*, which computes directionality instead of just similarity, is the Kullback–Leibler (KL) divergence from P to Q , measuring the information lost when approximating P with Q as $D_{\text{KL}}(P \| Q) = \int_{\mathbb{R}^d} P(x) \log \frac{P(x)}{Q(x)} dx$. For Gaussian densities

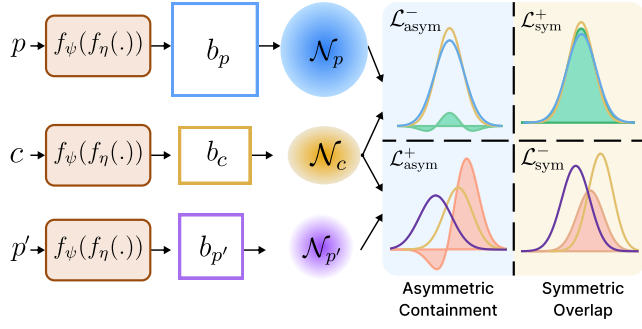


Figure 2: Overview of TaxoBell. Entities are encoded with $f_\eta(\cdot)$, mapped to axis-aligned boxes using $f_\psi(\cdot)$, and then projected to Gaussian embeddings. Training optimizes two energies on the Gaussians – a symmetric overlap term (BC) and an asymmetric containment term (KL).

P and Q , this admits the closed form,

$$D_{KL}(P \| Q) = \frac{1}{2} \left[\text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_2 - \mu_1)^\top \Sigma_2^{-1} (\mu_2 - \mu_1) - d + \log \frac{\det \Sigma_2}{\det \Sigma_1} \right]. \quad (4)$$

Low $D_{KL}(P \| Q)$ means that P is (softly) contained in Q , while large values signal poor coverage.

These energies act directly on μ and Σ , yield smooth gradients, and, under diagonal covariances, are inexpensive to compute, making them well suited for training Gaussian embeddings with both similarity (overlap) and hierarchy (containment) signals.

4 The TaxoBell Framework

4.1 Overview

In TaxoBell (Fig. 2), entities are encoded from text into axis-aligned boxes (Sec. 4.2) and then mapped to multivariate Gaussians (Sec. 4.3). Using the seed taxonomy for self-supervision, we train these Gaussians with two energies that (i) maximize symmetric parent–child overlap and (ii) enforce asymmetric hierarchical containment (Sec. 4.4).

4.2 Box Projection

Taxonomy nodes are conceptual entities, which are generally represented by surface names and definitions in natural language. To obtain a semantic vector for each entity e , we use a pre-trained language model $f_\eta(\cdot)$ (BERT [9] in our case). We construct the input sequence by concatenating the surface name and its definition with the BERT delimiter tokens “[CLS]” and “[SEP]” as, $x = [\text{CLS}] \ e \ [\text{SEP}] \ s \ [\text{SEP}]$ where s denotes the definition of the entity with surface name e . The encoder produces contextual token representations of “[CLS]”, “[SEP]”, e and s . Therefore, we take the final-layer [CLS] embedding as the representation of node e as, $\mathbf{n} = f_\eta(x)[0] \in \mathbb{R}^k$, where index 0 represents the position of embedding of [CLS] token while k is the embedding size.

Next, we project the entity representation n to a box $b = (c, o)$, where center $\mathbf{c} \in \mathbb{R}$ and offset $\mathbf{o} \in \mathbb{R}^{d^+}$ are vector embeddings with dimension d encoder $f_\psi(\cdot)$. Concretely, we use two independent 2-layered multilayer perceptrons (MLPs), namely MLP_c and MLP_o ,

to map to the center and offset, respectively, as,

$$c = \text{MLP}_c(n) = W_c^{(2)} \sigma(W_c^{(1)} n + b_c^{(1)}) + b_c^{(2)}, \quad (5)$$

$$o = \phi(\text{MLP}_o(n)) = \phi(W_o^{(2)} \sigma(W_o^{(1)} n + b_o^{(1)}) + b_o^{(2)}), \quad (6)$$

where $\sigma(\cdot)$ is a pointwise nonlinearity (e.g., ReLU or GELU) and W_c^i and W_o^i are weights corresponding to layers of MLPs of center and offset projectors, respectively. To ensure a valid box with strictly positive offsets in every dimension, we apply a monotone positivity function ϕ , such as exponential or softplus, to the offset. This gives us boxes $b_p = (c_p, o_p)$, $b_{p'} = (c_{p'}, o_{p'})$ and $b_c = (c_c, o_c)$ for parent, negative parent and child respectively.

4.3 Gaussian Projection

To enable boxes to model calibrated uncertainty, we project them to a multivariate Gaussian distribution $g = (\mu, \Sigma)$, where μ is the mean, while Σ is the covariance matrix. Intuitively, the box center c already locates the concept in space, so we identify it with the Gaussian mean, $\mu = c$. The offset governs dispersion and thus induces axis-aligned uncertainty. Concretely, we construct a positive semi-definite covariance by squaring the offsets and placing them on the diagonal as $\Sigma = \text{diag}(\mathbf{o} \odot \mathbf{o})$, $\Sigma_{ij} = o_i^2 \delta_{ij}$.

In particular, we obtain three Gaussian distributions, $\mathcal{N}_p(\mu_p, \Sigma_p)$, $\mathcal{N}_{p'}(\mu_{p'}, \Sigma_{p'})$ and $\mathcal{N}_c(\mu_c, \Sigma_c)$, for parent, negative parent and child respectively. This “box-to-Gaussian” projection of the parent preserves the geometric role of offsets, where a large offset means a wider box face and, correspondingly, a larger marginal variance. The induced equal-density contours of $\mathcal{N}(\mu, \Sigma)$ are axis-aligned hyper-ellipsoids nested around μ ; their sizes at 1/2/3 standard deviations provide an interpretable notion of semantic scope (e.g., the 68–95–99.7 rule), which a hard box cannot express. Moreover, this diagonal assumption is computationally efficient while still capturing concept generality through anisotropic variances. It also creates a tight link between the box “volume” and Gaussian spread: $\det(\Sigma) = \prod_{i=1}^d o_i^2$, so $\frac{1}{2} \log \det \Sigma = \sum_{i=1}^d \log o_i$ can be used as a differentiable proxy for semantic volume to prevent the problem of vanishing gradients. Crucially, moving from hard boxes to Gaussians replaces brittle, piecewise geometry with smooth overlap, enabling stable optimization with probabilistic objectives. Probabilistic containment of a child inside a parent is no longer a binary event but a graded relation governed jointly by mean separation and relative covariance, which our losses can shape continuously.

4.4 Gaussian Box Training and Inference

Self-Supervised Generation. We now optimize the Gaussian box embeddings so that they faithfully encode taxonomic hierarchy, i.e., asymmetric child–parent relations, under a self-supervised paradigm. The seed taxonomy $\mathcal{T}^0 = (\mathcal{N}^0, \mathcal{E}^0)$ provides natural supervision where every (child, anchor) edge is treated as a positive pair for training, where child node n_c is the “query” and parent node n_p is the “anchor”. To shape fine-grained decision boundaries, we mine N hard negatives ($\{n_{p'_i}\}_{i=1}^N \subset \mathcal{N}^0$) from the local neighborhood of the query node n_c , which are its siblings, uncles, cousins and grandparents. These negatives are topologically close to the child and semantically confusable with the parent, thus providing informative contrast to create distinctive boundaries. These $N+1$

pairs (one positive and N negatives) constitute a single training instance $\mathbf{X}_{\langle n_p, n_c \rangle} = \{\langle n_c, n_p, n_{p'_1} \rangle, \langle n_c, n_p, n_{p'_2} \rangle, \dots, \langle n_c, n_p, n_{p'_N} \rangle\}$. Repeating this procedure for every edge in $\mathcal{E}^0$ yields the full self-supervision set $\mathcal{X} = \{\mathbf{X}_{\langle n_p, n_c \rangle} \mid \langle n_p, n_c \rangle \in \mathcal{E}^0\}$. During training, each triple $\langle n_c, n_p, n_{p'_i} \rangle$ is encoded and then projected into a box and a Gaussian, which is optimized using energy-based optimization so that the positive Gaussian distributions overlap.

Energy-Based Optimization. We optimize the Gaussian boxes end-to-end by combining two complementary criteria: *symmetric overlap*, which captures conceptual similarity, and an *asymmetric* term that encodes hierarchical directionality. In addition, we apply regularization to prevent collapse to a point mass and to keep covariances well-conditioned. The overall objective aggregates the following components.

1. *Symmetric Overlap.* To capture direction-agnostic semantic similarity between a child n_c and its parent n_p , we use the Bhattacharyya coefficient $BC(\mathcal{N}_p, \mathcal{N}_c) = \exp(-D_B(\mathcal{N}_p, \mathcal{N}_c))$. BC is high when means are close and covariances are compatible, and it decreases as centers separate or uncertainty profiles diverge. The symmetric overlap loss for a triple $\langle n_c, n_p, n_{p'_i} \rangle$ is computed using binary cross entropy as,

$$\mathcal{L}_{\text{sym}} = -\log(BC(\mathcal{N}_p, \mathcal{N}_c)) - \log(1 - BC(\mathcal{N}_{p'}, \mathcal{N}_c)) \quad (7)$$

2. *Asymmetric Overlap.* To encode hierarchical directionality, we align the child distribution to its parent using the KL divergence $D_{\text{KL}}(\mathcal{N}_c \parallel \mathcal{N}_p)$, which quantifies the information lost when $\mathcal{N}_p$ approximates $\mathcal{N}_c$. If the child’s mass is well contained within the parent, the value of $D_{\text{KL}}(\mathcal{N}_c \parallel \mathcal{N}_p)$ is small compared to negative pairs. We contrast this against a hard negative parent $\mathcal{N}_{p'}$ with a margin-based triplet objective:

$$\mathcal{L}_{\text{align}} = \max(0, D_{\text{KL}}(\mathcal{N}_c \parallel \mathcal{N}_p) - D_{\text{KL}}(\mathcal{N}_c \parallel \mathcal{N}_{p'}) + \delta), \quad (8)$$

where $\delta > 0$ enforces a minimum separation between the positive and negative pairs.

However, just enforcing $\mathcal{L}_{\text{align}}$ does not control the coverage of the parent because a narrowly peaked parent may still yield a small KL, also reducing the child to a small peak. To ensure parents remain broader than their children in order to accommodate more children, we introduce a reverse-KL term that is coupled to the log-volume gap, $D_{\text{KL}}(\mathcal{N}_p \parallel \mathcal{N}_c) \geq \log \text{Vol}(\mathcal{N}_p) - \log \text{Vol}(\mathcal{N}_c) = D_{\text{rep}}$. We penalize violations of the coverage constraint with a hinge, as

$$\mathcal{L}_{\text{diverge}} = \max\{0, C \times D_{\text{rep}} - D_{\text{KL}}(\mathcal{N}_p \parallel \mathcal{N}_c)\}, \quad (9)$$

where $C > 0$ scales the required informational separation by the geometric (volume) separation. The final asymmetric objective is a weighted combination,

$$\mathcal{L}_{\text{asym}} = \mathcal{L}_{\text{align}} + \lambda \mathcal{L}_{\text{diverge}}, \quad (10)$$

with hyperparameter $\lambda > 0$, yielding stable training and consistent hierarchical alignment. We discuss the geometrical implications of eq.9 and 10 for different values of C and λ and their impact on model performance in Appendix F.

3. *Volume Regularization.* To prevent numerical instability and avoid degenerate Gaussians, we introduce two regularizers. The first regularizer, *minimum volume regularization*, prevents volume collapse by enforcing a minimum standard deviation. The minimum regularization loss for an entity distribution $\mathcal{N}_x(\mu_x, \Sigma_x)$ is formulated as a squared hinge loss,

$$\mathcal{L}_{\text{reg}}(\mathcal{N}_x) = \frac{1}{d} \left\| (\delta_{\text{var}} I - \Sigma_x)_+ \right\|_F^2, \quad (11)$$

where $\delta_{\text{var}} > 0$ is the minimum variance, I is an identity matrix of dimension d , $(\cdot)_+$ denotes the element-wise hinge such that $(A_{ij})_+ = \max(0, A_{ij})$ while $\|\cdot\|_F$ is the Frobenius norm.

The second regularizer, *covariance clipping*, enhances stability by preventing variances from becoming excessively large, which can lead to floating-point errors. This term penalizes variances that exceed a maximum threshold, M_{var} ,

$$\mathcal{L}_{\text{clip}}(\mathcal{N}_x) = \frac{1}{d} \text{tr}([\Sigma_x - M_{\text{var}} I]_+), \quad (12)$$

where $[A]_+$ denotes the elementwise hinge $\max(0, A_{ij})$, $\text{tr}(\cdot)$ is the trace, I is the identity, d is the embedding dimension, and $M_{\text{var}} > 0$ is the variance ceiling. These volume regularizations are computed for parent, child, and negative parent distributions ($\mathcal{N}_p, \mathcal{N}_c, \mathcal{N}_{p'}$).

Therefore, we combine the above objective functions as follows,

$$\mathcal{L}_{\text{overall}} = \mathcal{L}_{\text{sym}} + \mathcal{L}_{\text{asym}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{clip}} \quad (13)$$

Inference. During inference, the goal is to attach a new query concept to an appropriate parent (anchor) in the seed taxonomy. Given a query q , we encode it with the pretrained encoder, project it to a box $b_q = (c_q, o_q)$, and obtain its Gaussian embedding $\mathcal{N}_q(\mu_q, \Sigma_q)$ (Sec. 4.2, 4.3). For each candidate anchor a in the seed taxonomy, with Gaussian $\mathcal{N}_a(\mu_a, \Sigma_a)$, we compute two scores – a *symmetric overlap* term via the Bhattacharyya coefficient $BC(\mathcal{N}_a, \mathcal{N}_q)$, and an *asymmetric containment* term via the KL $D_{\text{KL}}(\mathcal{N}_q \parallel \mathcal{N}_a)$ separately, naming two models $\text{TaxoBell}_{\text{BC}}$ and $\text{TaxoBell}_{\text{KL}}$. For visualization or downstream rules, the chosen Gaussian can be translated back to a box by fixing a confidence level (e.g., $1\sigma, 2\sigma$) and mapping per-dimension spreads to offsets, preserving the uncertainty.

Gaussian to Box Translation. To render a multivariate Gaussian $\mathcal{N}(\mu, \Sigma)$ as an axis-aligned box $b = (c, o)$, we set the center to the mean $c = \mu$ and the offset to $o = k \sqrt{\Sigma}$. Choosing $k \in \{1, 2, 3\}$ gives roughly 68%, 95%, and 99.7% coverage per dimension, respectively.

5 Experiments

We evaluate TaxoBell on five real-world taxonomies against vector- and geometry-based baselines using standard ranking/retrieval metrics. The implementation details are in Appendix A.

5.1 Experimental Setup

5.1.1 *Datasets.* Following [14, 22, 42], we evaluate TaxoBell on five public taxonomies spanning general and specialized domains for comprehensive coverage across major domains – (i) Environment (ENV), (ii) Science (SCI) from SemEval-2016 Task 13 [5], (iii) Food from SemEval-2015 Task 17 [4], (iv) WordNet sub-taxonomies from Bansal et al. [3], and (v) Medical Subject Headings (MeSH), a widely used clinical taxonomy [21], where (i), (ii) and (iv) are single parent taxonomies while others are multi-parent ones. Full dataset statistics and train-test split are discussed in Appendix B.

Table 1: Performance comparison of TaxoBell with baseline methods. Each entry in the table reports mean^{std-dev} in percentage over five independent runs with distinct random seeds. The best performance is marked in bold, while the best baseline is underlined. For MR (not a percentage), lower values indicate better performance and are marked with “↓”.

Methods	Science					Environment					WordNet				
	MR↓	MRR	R@1	R@5	Wu&P	MR↓	MRR	R@1	R@5	Wu&P	MR↓	MRR	R@1	R@5	Wu&P
BERT+MLP	241.6 ^{23.1}	21.3 ^{3.7}	13.1 ^{4.1}	27.3 ^{3.3}	45.7 ^{1.7}	74.2 ^{7.2}	21.4 ^{2.6}	11.1 ^{3.0}	31.8 ^{2.1}	48.2 ^{0.4}	314.6 ^{62.2}	16.8 ^{2.1}	9.1 ^{2.5}	38.1 ^{3.7}	41.4 ^{1.2}
TaxoExpan	117.6 ^{15.7}	40.1 ^{2.8}	24.6 ^{3.9}	41.8 ^{3.1}	55.7 ^{1.2}	56.1 ^{5.3}	28.4 ^{3.1}	12.9 ^{5.7}	37.1 ^{2.3}	49.8 ^{1.0}	141.6 ^{17.3}	31.1 ^{2.2}	17.1 ^{3.5}	38.1 ^{1.6}	49.7 ^{1.8}
Arborist	83.1 ^{7.2}	41.2 ^{3.1}	26.5 ^{4.4}	51.5 ^{3.5}	61.2 ^{1.4}	39.1 ^{5.8}	33.7 ^{4.7}	24.9 ^{5.9}	45.2 ^{2.1}	52.5 ^{1.3}	92.6 ^{3.6}	33.7 ^{2.7}	20.3 ^{3.5}	43.9 ^{2.8}	53.5 ^{1.3}
BoxTaxo	67.7 ^{11.3}	43.0 ^{3.8}	25.3 ^{4.5}	49.2 ^{3.1}	63.1 ^{1.7}	34.1 ^{7.3}	41.6 ^{4.9}	32.3 ^{6.2}	51.4 ^{3.5}	65.1 ^{1.4}	77.1 ^{8.8}	34.1 ^{3.2}	22.3 ^{4.2}	45.7 ^{3.6}	58.1 ^{1.7}
TMN	54.2 ^{5.1}	45.5 ^{2.5}	31.5 ^{3.8}	53.7 ^{1.9}	65.7 ^{1.2}	31.3 ^{3.5}	43.8 ^{2.1}	34.7 ^{3.7}	51.1 ^{3.2}	63.9 ^{2.0}	73.9 ^{4.9}	35.8 ^{2.7}	23.7 ^{3.2}	49.0 ^{2.6}	56.6 ^{0.8}
STEAM	31.7 ^{3.3}	50.7 ^{3.5}	34.8 ^{4.9}	59.1 ^{4.1}	72.2 ^{1.3}	27.1 ^{2.8}	44.2 ^{2.7}	34.1 ^{3.4}	55.6 ^{2.9}	65.2 ^{1.7}	61.1 ^{2.4}	37.3 ^{2.1}	24.9 ^{3.9}	54.5 ^{1.7}	59.2 ^{1.2}
TaxoEnrich	22.1 ^{3.8}	55.2 ^{4.9}	39.5 ^{5.1}	65.4 ^{3.1}	72.7 ^{2.2}	17.2 ^{6.1}	49.8 ^{4.3}	42.1 ^{5.3}	61.0 ^{1.2}	71.8 ^{2.5}	54.3 ^{3.5}	45.1 ^{3.1}	27.2 ^{4.3}	61.7 ^{4.3}	68.7 ^{1.9}
TaxoBell_{BC}	13.2^{0.9}	58.2^{1.2}	49.0^{1.9}	74.8^{1.3}	76.2^{0.4}	8.3^{0.5}	58.7^{2.5}	46.5^{4.4}	75.0^{1.4}	77.2^{1.0}	40.9^{4.2}	50.3^{1.4}	36.1^{0.8}	70.3^{2.4}	73.4^{0.3}
TaxoBell_{KL}	13.8^{1.2}	58.5^{1.6}	48.0^{2.8}	74.1^{0.6}	75.6^{0.9}	8.9^{0.7}	58.4^{1.7}	47.3^{3.8}	75.0^{1.3}	75.8^{0.5}	41.0^{4.2}	51.2^{1.2}	37.1^{0.6}	70.4^{2.1}	73.5^{0.8}

Methods	SemEval-Food						MeSH					
	MR↓	MRR	R@1	R@5	H@1	H@5	MR↓	MRR	R@1	R@5	H@1	H@5
BERT+MLP	508.8 ^{35.3}	47.1 ^{2.1}	12.3 ^{2.7}	26.1 ^{2.9}	13.8 ^{1.2}	25.7 ^{0.2}	1352.4 ^{57.1}	9.3 ^{1.7}	1.1 ^{0.6}	5.2 ^{0.9}	3.7 ^{0.3}	7.2 ^{0.3}
TaxoExpan	343.8 ^{17.2}	41.3 ^{1.7}	25.1 ^{2.2}	32.8 ^{2.3}	14.2 ^{0.7}	27.3 ^{0.9}	891.1 ^{31.4}	17.1 ^{2.1}	2.9 ^{0.8}	10.3 ^{0.9}	7.3 ^{0.5}	15.8 ^{0.1}
BoxTaxo	363.2 ^{12.8}	45.2 ^{1.6}	27.3 ^{2.3}	36.4 ^{2.1}	29.1 ^{1.8}	41.1 ^{0.5}	620.2 ^{22.9}	21.5 ^{2.9}	17.1 ^{1.1}	30.2 ^{0.6}	16.5 ^{1.2}	31.7 ^{1.1}
Arborist	247.9 ^{21.1}	45.3 ^{1.9}	29.3 ^{2.1}	37.7 ^{2.2}	21.5 ^{2.7}	37.2 ^{1.4}	553.6 ^{26.2}	21.3 ^{2.3}	19.0 ^{1.7}	31.1 ^{1.3}	16.7 ^{1.4}	29.4 ^{0.9}
TMN	192.6 ^{16.1}	51.8 ^{2.1}	34.1 ^{2.6}	43.2 ^{1.9}	25.3 ^{2.1}	40.9 ^{1.1}	433.7 ^{16.5}	23.5 ^{1.2}	18.1 ^{0.6}	33.2 ^{0.9}	16.6 ^{1.8}	31.8 ^{1.0}
STEAM	155.9 ^{14.0}	53.8 ^{2.5}	39.1 ^{3.1}	51.3 ^{1.7}	34.4 ^{2.3}	58.7 ^{0.4}	372.6 ^{16.2}	25.1 ^{3.1}	17.7 ^{2.7}	35.1 ^{1.0}	18.2 ^{1.8}	38.2 ^{1.5}
TaxoEnrich	101.7 ^{11.2}	53.7 ^{3.3}	41.8 ^{2.7}	57.5 ^{1.1}	39.2 ^{1.9}	67.2 ^{1.2}	247.7 ^{15.5}	25.3 ^{2.7}	18.7 ^{2.1}	37.6 ^{1.3}	21.3 ^{1.9}	47.1 ^{1.2}
TaxoBell_{BC}	36.3^{7.3}	58.1^{1.7}	46.0^{2.1}	76.0^{0.4}	45.1^{1.7}	75.2^{0.8}	175.4^{4.7}	28.9^{1.9}	20.2^{2.8}	42.0^{1.1}	24.5^{1.7}	54.3^{1.1}
TaxoBell_{KL}	36.6^{6.7}	57.4^{2.0}	44.7^{3.4}	76.1^{1.6}	45.1^{2.5}	75.6^{1.1}	175.4^{9.1}	29.2^{2.2}	20.1^{3.1}	42.3^{1.5}	24.8^{2.4}	55.3^{1.9}

Table 2: Combined Statistical Test Results using Fisher’s Method comparing TaxoBell to the best baseline.

	SCI	ENV	WordNet
χ^2	90.71	73.01	100.62
p -value	1.34×10^{-15}	7.56×10^{-12}	9.61×10^{-18}

	SemEval-Food	MeSH
χ^2	85.66	62.38
p -value	8.53×10^{-13}	1.58×10^{-8}

Table 3: Impact of the asymmetric optimization ($\mathcal{L}_{\text{asym}}$) on SCI, MeSH, and Food benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Science			Food			MeSH		
	H@1	MR↓	MRR	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O $\mathcal{L}_{\text{asym}}$	42.35	13.44	55.07	33.89	49.39	48.51	22.70	249.32	25.86
$\mathcal{L}_{\text{asym}}$	51.70	12.77	58.50	46.93	36.60	59.44	26.50	175.35	29.15
↑%	+22.08	+4.99	+6.23	+38.48	+25.90	+22.53	+16.74	+29.67	+12.72

5.1.2 Baselines. We aim to study the effectiveness of Gaussian box embeddings over vector and geometric embeddings for representing and expanding taxonomic hierarchies. We evaluate TaxoBell against seven baselines spanning vector and structure-aware approaches: (i) BERT+MLP [30], (ii) Arborist [25], (iii) TaxoExpan

Table 4: Impact of the symmetric optimization ($\mathcal{L}_{\text{sym}}$) on SCI, MeSH, and Food benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Science			Food			MeSH		
	H@1	MR↓	MRR	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O $\mathcal{L}_{\text{sym}}$	11.77	196.57	5.75	25.77	75.67	39.26	11.24	322.75	18.89
$\mathcal{L}_{\text{sym}}$	50.91	14.09	59.41	43.43	29.94	56.35	23.18	171.76	27.99
↑%	+332.5	+92.8	+933.2	+68.5	+60.4	+43.5	+106.3	+46.8	+48.2

[33], (iv) TMN [45], (v) STEAM [44], (vi) BoxTaxo [15], and (vii) TaxoEnrich [14]. The baselines are discussed in Appendix C.

5.1.3 Evaluation Metrics. Following prior works [28, 40, 42], for each query node q , we rank all candidate anchors in the seed taxonomy and record the rank(s) of the gold parent(s). We report **Mean Rank (MR)**, **Mean Reciprocal Rank (MRR)**, **Recall@ k** , **Hit@ k** and **Wu & Palmer**. Metric definitions and the choices for single and multi-parent taxonomies are detailed in Appendix D.

5.2 Comparative Analysis and Statistical Tests

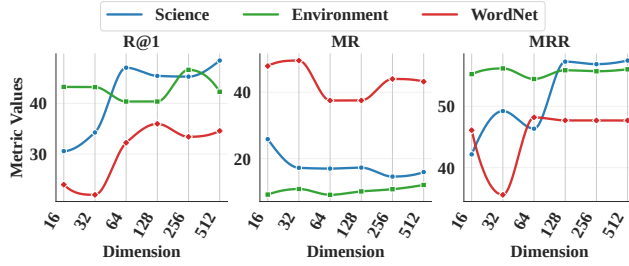
Table 1 presents the results of all baselines compared with two variants of our method – TaxoBell_{KL}, with negative KL divergence as ranker, and TaxoBell_{BC} with Bhattacharyya coefficient as ranker. Vector embedding methods, such as BERT+MLP, perform the worst, indicating that symmetric similarity in a point space is ill-suited to hierarchical attachment, while path-based methods such

Table 5: Effect of the volume regularization ($\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{clip}}$) on SCI, MeSH, and Food benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Science			Food			MeSH		
	H@1	MR↓	MRR	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O Reg	43.52	20.14	50.34	39.83	47.55	46.05	22.92	198.61	26.16
Reg	51.70	12.77	58.50	46.93	36.60	59.44	26.50	175.35	29.15
↑ %	+18.80	+36.59	+16.21	+17.83	+23.03	+29.08	+15.62	+11.71	+11.43

Table 6: Effect of $\mathcal{L}_{\text{diverge}}$ on SCI, MeSH, and Food benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Science			Food			MeSH		
	H@1	MR↓	MRR	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O $\mathcal{L}_{\text{diverge}}$	45.88	24.81	55.86	38.93	43.77	50.25	23.02	214.14	26.21
$\mathcal{L}_{\text{diverge}}$	51.70	12.77	58.50	46.93	36.60	59.44	26.50	175.35	29.15
↑ %	+12.69	+48.53	+4.73	+20.55	+16.38	+18.29	+15.12	+18.11	+11.22

**Figure 3: Effect of embedding dimensionality on performance for SCI, ENV, and WordNet benchmarks.**

as TaxoExpan, Arborist, STEAM, and TaxoEnrich improve over point baselines but still lag behind our Gaussian-box models. Both TaxoBell_{KL} and TaxoBell_{BC} consistently achieve the lowest MR and the highest MRR/Recall across datasets, with the largest gains on multi-parent taxonomies. Between our variants, TaxoBell_{KL} strongest R@1, reflecting sharper asymmetric containment, while TaxoBell_{BC} often matches or slightly improves MR/MRR, suggesting smoother overlap. In sum, representing concepts as Gaussian boxes, retaining box geometry while modeling per-dimension uncertainty, yields state-of-the-art ranking quality on both single- and multi-parent benchmarks.

Statistical tests. We compare TaxoBell to the strongest baseline (TaxoEnrich) and, per dataset, combine metric-level p -values using Fisher’s method, which under the null follows a χ^2 distribution with $2k$ degrees of freedom, where k is the number of metrics. Table 2 reports both X^2 and the combined p -value, showing large χ^2 values and small p -values on every dataset, so we reject the null hypothesis that TaxoBell and the best baseline perform equivalently and conclude that TaxoBell’s gains are statistically significant.

Table 7: Direct vs. Gaussian projection variants of TaxoBell on SCI and ENV benchmarks. D_* is vector embedding to Gaussian projection while G_* is box-to-Gaussian projection. For MR, lower values indicate better performance.

Methods	Science			Environment		
	MR↓	MRR	R@1	MR↓	MRR	R@1
TaxoBell _{D_{KL}}	58.10	18.83	5.82	44.62	25.50	19.23
TaxoBell _{G_{KL}}	13.96	58.50	51.76	8.98	58.36	47.31
↑ %	+76.0	+210.7	+789.3	+79.9	+128.9	+146.0
TaxoBell _{D_{BC}}	57.76	18.95	11.76	44.67	24.48	19.23
TaxoBell _{G_{BC}}	13.18	58.22	48.94	8.31	58.68	46.54
↑ %	+77.2	+207.2	+316.2	+81.4	+139.7	+142.0

5.3 Ablation Studies

We ablate TaxoBell on SCI, Food, and MeSH to isolate component effects and compare alternatives. Additional ablations and hyperparameter studies are in Appendices E and F, respectively.

5.3.1 Impact of Asymmetric and Symmetric Losses. We study the effect of both complementary losses, $\mathcal{L}_{\text{asym}}$ and $\mathcal{L}_{\text{sym}}$, on the performance of TaxoBell as shown in Tables 3 and 4. Removing $\mathcal{L}_{\text{asym}}$ erases the notion of directionality, so queries cluster near related anchors but attach to structurally wrong nodes such as siblings, uncles, or overly general ancestors, yielding higher MR and lower R@1/MRR. Furthermore, dropping $\mathcal{L}_{\text{sym}}$ leaves the model only with the KL optimization, which can be satisfied by shrinking the child or inflating the parent, degrading semantic geometry and calibration – again hurting MR and R@1/MRR. Using both losses jointly gives calibrated, non-degenerate embeddings and the strongest ranks, showing hierarchy and similarity are complementary signals for reliable attachment.

5.3.2 Impact of Volume Regularization. We also study the effect of volume regularization, consisting of two components, the minimum-variance regularizer $\mathcal{L}_{\text{reg}}$ and the variance clipping term $\mathcal{L}_{\text{clip}}$, on model performance as shown in Table 5. We observe that removing these terms consistently hurts performance across datasets and metrics. Without a lower bound on variance, the model can shrink a child’s covariance to satisfy KL, making parent and child nearly indistinguishable and destroying fine-grained ordering. Without an upper bound, it can inflate a parent to fake the containment, yielding overly broad ellipsoids that break hierarchical margins and destabilize training. Enforcing both bounds keeps covariances well-conditioned, yields calibrated uncertainty, preserves the intended relation, and produces more reliable containment decisions.

5.3.3 Impact of Reverse KL Divergence. We study the impact of reverse-KL loss, $\mathcal{L}_{\text{diverge}}$ on TaxoBell’s performance in Table 6. Using only the forward KL alignment lets the model minimize containment by making the parent narrowly peaked (or shrinking the child), producing brittle, poorly calibrated embeddings. Adding reverse KL imposes a coverage margin, stabilizes covariances, and improves separation among siblings. Empirically, this consistently lowers MR and raises H@1/MRR, acting as an uncertainty-margin

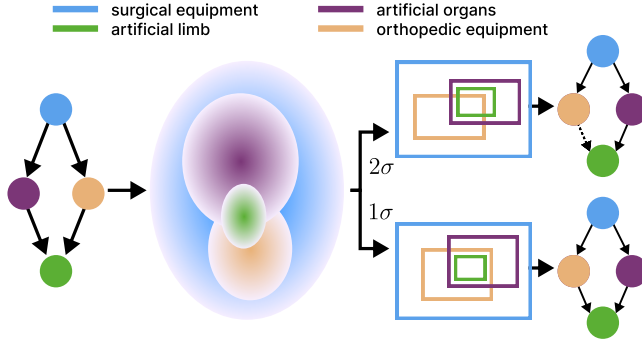


Figure 4: Case study of reconstructing a MeSH multi-parent subtree from learned Gaussians by controlling the uncertainty while converting Gaussians into boxes.

regularizer that complements forward KL to better respect both hierarchy and semantics.

5.3.4 Direct Projection vs Gaussian Projection. We further study the role of geometry in the projection as shown in Table 7. In a *direct* variant, we map the encoder output $f_\psi(\cdot)$ straight to a Gaussian by using the hidden vector as μ and deriving a diagonal Σ from its magnitudes. This performs worse on all benchmarks because uncertainty is tied to the encoder’s feature basis, so the model cannot easily widen parents or separate siblings along the most useful directions. Our TaxoBell projection instead maps each concept to an axis-aligned box first and then converts that box into a Gaussian. This keeps a clear geometric bias (containment), provides interpretable variances, and lets training directly control spread to satisfy energy-based losses. The result is calibrated, non-degenerate Gaussians that capture asymmetric parent–child relations more faithfully and yield stronger rankings.

5.4 Dimensionality Analysis

To study how embedding size affects taxonomy expansion, we vary the dimension of the Gaussian boxes and plot the metrics in Fig. 3. We observe that varying the embedding size reveals the expected capacity trade-off rather than a single monotone trend. Specifically on SCI, increasing the dimensions generally helps once the model has enough degrees of freedom to separate siblings and encode containment. ENV is more volatile as gains at smaller dimensions give way to a dip and only partial recovery at larger sizes, indicating sensitivity to over-parameterization relative to the data scale. WordNet saturates quickly, and moving from very small to moderate dimensions brings most of the improvement, after which the curves flatten. Overall, very small dimensions underfit, very large dimensions can overfit or destabilize covariance learning, and a mid-range dimensionality provides the best accuracy–stability balance.

5.5 Case Study and Error Analysis

Fig. 4 illustrates an example of converting the MeSH subtree’s Gaussians into boxes by choosing the confidence level. Using 1σ , we get tighter boxes with lower uncertainty, cleanly capturing the intended multi-parent links, while with 2σ , the boxes grow in size,

Table 8: Qualitative examples of TaxoBell on the Science and WordNet benchmarks. For each benchmark, we show one correct and one incorrect attachment. Correct predictions are marked with ✓ while incorrect with ✗.

Query	Anchor (Gold)	Top Predictions
Science		
Solid Geometry	Geometry	Geometry ✓
		Spherical geometry ✗
		Plane geometry ✗
Orthopedics	Medical Science	Rheumatology ✗
		Medicine ✗
		Traumatology ✗
WordNet		
Convection	Temperature change	Temperature change ✓
		Climate change ✗
		Advection ✗
Infanticide	Murderer	low-birth-weight baby ✗
		matricide ✗
		kittee ✗

increasing overlap and the child region can extend beyond the true shared intersection of its parents, which leads to confusion whether the child is multi-parent or misclassified edge due to erroneous intersection (shown by dashed arrow connecting *orthopedic equipment* to *artificial limb*). Furthermore, Table 8 complements this with single-parent examples from Science and WordNet. Typical successes include *Solid geometry* placed under *Geometry* in SCI because the definition and usage strongly match, which yields high symmetric overlap and *Convection* placed under *Temperature change* in WordNet since the query emphasizes heat transfer and temperature variation, so its mass sits well inside the anchor while remaining far from distractors like *Climate change* or *Heating*. We also observe two representative errors. In Science, *Orthopedics* is attached to *Rheumatology* instead of the gold *Medical Science*, reflecting sibling ambiguity when several nearby anchors are semantically close. In WordNet, *Infanticide* drifts toward unrelated or overly broad anchors, indicating weak domain grounding. Further case studies on multi-parent taxonomies are discussed in Appendix G.

6 Conclusion

In this paper, we introduce TaxoBell, a self-supervised framework for taxonomy expansion that represents concepts as Gaussian boxes and optimizes them using two complementary energies, namely a symmetric Bhattacharyya overlap and an asymmetric KL-based hierarchical containment, resulting in a stable training recipe with volume regularization, which yields calibrated, non-degenerate embeddings that respect hierarchy while preserving semantic proximity. Across multiple taxonomy expansion benchmarks (single- and multi-parent), TaxoBell consistently reduces MR and improves MRR/Recall over strong vector, path-based, and hard-box baselines, and ablations verify that both losses, the learned Gaussian projection, and volume control are necessary for the gains. The representation is interpretable, probabilistic, and readily extensible. We believe probabilistic geometric representations of this kind offer a practical foundation for robust, scalable hierarchical reasoning in real-world systems.

References

- [1] Ben Athiwaratkun, Andrew Wilson, and Anima Anandkumar. 2018. Probabilistic FastText for Multi-Sense Word Embeddings. In *Proceedings of ACL*. 1–11.
- [2] Ben Athiwaratkun and Andrew Gordon Wilson. 2017. Multimodal word distributions. *arXiv preprint arXiv:1704.08424* (2017).
- [3] Mohit Bansal, David Burkett, Gerard de Melo, and Dan Klein. 2014. Structured Learning for Taxonomy Induction with Belief Propagation. In *Proceedings of ACL*. 1041–1051.
- [4] Georgeta Bordea, Paul Buitelaar, Stefano Faralli, Roberto Navigli, et al. 2015. Semeval-2015 task 17: Taxonomy extraction evaluation (texeval). In *Proceedings of SemEval*. 902–910.
- [5] Georgeta Bordea, Els Lefever, and Paul Buitelaar. 2016. SemEval-2016 Task 13: Taxonomy Extraction Evaluation (TExEval-2). In *Proceedings of SemEval*. 1081–1091.
- [6] Haw-Shiuan Chang, Ziyun Wang, Luke Vilnis, and Andrew McCallum. 2018. Distributional Inclusion Vector Embedding for Unsupervised Hypernymy Detection. In *Proceedings of NAACL*. 485–495.
- [7] Tejas Chheda, Purujit Goyal, Trang Tran, Dhruvesh Patel, Michael Boratko, Shib Sankar Dasgupta, and Andrew McCallum. 2021. Box Embeddings: An open-source library for representation learning using geometric structures. In *Proceedings of EMNLP*. 203–211.
- [8] Shib Dasgupta, Michael Boratko, Dongxu Zhang, Luke Vilnis, Xiang Li, and Andrew McCallum. 2020. Improving Local Identifiability in Probabilistic Box Embeddings. In *NeurIPS*, Vol. 33. 182–192.
- [9] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL*. 4171–4186.
- [10] Shizhu He, Kang Liu, Guoliang Ji, and Jun Zhao. 2015. Learning to Represent Knowledge Graphs with Gaussian Embedding. In *Proceedings of CIKM*. 623–632.
- [11] Marti A. Hearst. 1992. Automatic Acquisition of Hyponyms from Large Text Corpora. In *COLING 1992 Volume 2: The 14th International Conference on Computational Linguistics*.
- [12] Jiaxin Huang, Yiqing Xie, Yu Meng, Yunyi Zhang, and Jiawei Han. 2020. Corel: Seed-guided topical taxonomy construction by concept learning and relation transferring. In *Proceedings of KDD*. 1928–1936.
- [13] Tony Jebara, Risi Kondor, and Andrew Howard. 2004. Probability Product Kernels. *J. Mach. Learn. Res.* 5 (2004), 819–844.
- [14] Minhao Jiang, Xiangchen Song, Jieyu Zhang, and Jiawei Han. 2022. Taxoerich: Self-supervised taxonomy completion via structure-semantic representations. In *Proceedings of WWW*. 925–934.
- [15] Song Jiang, Qiyue Yao, Qifan Wang, and Yizhou Sun. 2023. A Single Vector Is Not Enough: Taxonomy Expansion via Box Embeddings. In *WWW*. 2467–2476.
- [16] Giannis Karamanolakis, Jun Ma, and Xin Luna Dong. 2020. TXtract: Taxonomy-Aware Knowledge Extraction for Thousands of Product Categories. In *Proceedings of ACL*. 8489–8502.
- [17] Jacob Devlin Ming-Wei Chang Kenton and Lee Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of naacl-HLT*, Vol. 1. Minneapolis, Minnesota, 2.
- [18] Yann LeCun, Sumit Chopra, Raia Hadsell, M Ranzato, Fufei Huang, et al. 2006. A tutorial on energy-based learning. *Predicting structured data* 1, 0 (2006).
- [19] Dongha Lee, Jiaming Shen, SeongKu Kang, Susik Yoon, Jiawei Han, and Hwanjo Yu. 2022. Taxocom: Topic taxonomy completion with hierarchical discovery of novel topic clusters. In *Proceedings of WWW*. 2819–2829.
- [20] Xiang Li, Luke Vilnis, Dongxu Zhang, Michael Boratko, and Andrew McCallum. 2019. Smoothing the Geometry of Probabilistic Box Embeddings. In *ICLR*.
- [21] Carolyn E Lipscomb. 2000. Medical subject headings (MeSH). *Bulletin of the Medical Library Association* (2000), 265.
- [22] Zichen Liu, Hongyuan Xu, Yanlong Wen, Ning Jiang, Haiying Wu, and Xiaojie Yuan. 2021. TEMP: taxonomy expansion with dynamic margin loss through taxonomy-paths. In *Proceedings of EMNLP*. 3854–3863.
- [23] Xusheng Luo, Luxin Liu, Yonghua Yang, Le Bo, Yuanpeng Cao, Jinghang Wu, Qiang Li, Keping Yang, and Kenny Q Zhu. 2020. AliCoCo: Alibaba e-commerce cognitive concept net. In *Proceedings of SIGMOD*. 313–327.
- [24] A. Mahabal, Jiyun Luo, Rui Huang, Michael Ellsworth, and Rui Li. 2023. Producing Usable Taxonomies Cheaply and Rapidly at Pinterest Using Discovered Dynamic μ -Topics. *ArXiv abs/2301.12520* (2023).
- [25] Emaad Manzoor, Rui Li, Dhananjay Shroutry, and Jure Leskovec. 2020. Expanding taxonomies with implicit edge semantics. In *Proceedings of WWW*. 2044–2054.
- [26] Yuning Mao, Tong Zhao, Andrey Kan, Chenwei Zhang, Xin Luna Dong, Christos Faloutsos, and Jiawei Han. 2020. Octet: Online catalog taxonomy enrichment with self-supervision. In *Proceedings of KDD*. 2247–2257.
- [27] Sahil Mishra, Kumar Arjun, and Tanmoy Chakraborty. 2025. Rank, Chunk and Expand: Lineage-Oriented Reasoning for Taxonomy Expansion. In *Findings of ACL*. 12935–12953.
- [28] Sahil Mishra, Avi Patni, Niladri Chatterjee, and Tanmoy Chakraborty. 2025. QuanTaxo: A Quantum Approach to Self-Supervised Taxonomy Expansion. *arXiv preprint arXiv:2501.14011* (2025).
- [29] Sahil Mishra, Ujjwal Sudev, and Tanmoy Chakraborty. 2024. FLAME: Self-Supervised Low-Resource Taxonomy Expansion Using Large Language Models. *ACM Trans. Intell. Syst. Technol.* (Dec. 2024).
- [30] Alexander Panchenko, Stefano Faralli, Eugen Ruppert, Steffen Remus, Hubert Naets, Cédric Faron, Simone Paolo Ponzetto, and Chris Biemann. 2016. A Taxonomy Induction Method based on Lexico-Syntactic Patterns, Substrings and Focused Crawling. In *Proceedings of SemEval*. 1320–1327.
- [31] Dhruvesh Patel, Shib Sankar Dasgupta, Michael Boratko, Xiang Li, Luke Vilnis, and Andrew McCallum. 2020. Representing Joint Hierarchies with Box Embeddings. In *Automated Knowledge Base Construction*.
- [32] Hongyu Ren, Weihua Hu, and Jure Leskovec. 2020. Query2box: Reasoning over Knowledge Graphs in Vector Space Using Box Embeddings. In *ICLR*.
- [33] Jiaming Shen, Zhihong Shen, Chenyan Xiong, Chi Wang, Kuansan Wang, and Jiawei Han. 2020. TaxoExpan: Self-supervised taxonomy expansion with position-enhanced graph neural network. In *Proceedings of WWW*. 486–497.
- [34] Jiaming Shen, Zeqiu Wu, Dongming Lei, Chao Zhang, Xiang Ren, Michelle T Vanni, Brian M Sadler, and Jiawei Han. 2018. Hiexpan: Task-guided taxonomy construction by hierarchical tree expansion. In *Proceedings of KDD*. 2180–2189.
- [35] Rion Snow, Daniel Jurafsky, and Andrew Ng. 2004. Learning Syntactic Patterns for Automatic Hypernym Discovery. In *NeurIPS*, Vol. 17.
- [36] Nikhita Vedula, Patrick K Nicholson, Deepak Ajwani, Sourav Dutta, Alessandra Sala, and Srinivasan Parthasarathy. 2018. Enriching taxonomies with functional domain knowledge. In *SIGIR*. 745–754.
- [37] Luke Vilnis, Xiang Li, Shikhar Murty, and Andrew McCallum. 2018. Probabilistic Embedding of Knowledge Graphs with Box Lattice Measures. In *Proceedings of ACL*. 263–272.
- [38] Luke Vilnis and Andrew McCallum. 2014. Word representations via gaussian embedding. *arXiv preprint arXiv:1412.6623* (2014).
- [39] Suyuchen Wang, Ruihui Zhao, Xi Chen, Yefeng Zheng, and Bang Liu. 2021. Enquire one’s parent and child before decision: Fully exploit hierarchical structure for self-supervised taxonomy expansion. In *Proceedings of WWW*. 3291–3304.
- [40] Suyuchen Wang, Ruihui Zhao, Yefeng Zheng, and Bang Liu. 2022. Qen: Applicable taxonomy completion via evaluating full taxonomic relations. In *Proceedings of WWW*. 1008–1017.
- [41] Hongyuan Xu, Yunong Chen, Zichen Liu, Yanlong Wen, and Xiaojie Yuan. 2022. TaxoPrompt: A Prompt-based Generation Method with Taxonomic Context for Self-Supervised Taxonomy Expansion. In *Proceedings of IJCAI*. 4432–4438. Main Track.
- [42] Hongyuan Xu, Ciyi Liu, Yuhang Niu, Yunong Chen, Xiangrui Cai, Yanlong Wen, and Xiaojie Yuan. 2023. TacoPrompt: A Collaborative Multi-Task Prompt Learning Method for Self-Supervised Taxonomy Completion. In *Proceedings of EMNLP*. 15804–15817.
- [43] Wei Xue, Yongliang Shen, Wenqi Ren, Jietian Guo, Shiliang Pu, and Weiming Lu. 2024. Insert or Attach: Taxonomy Completion via Box Embedding. In *Proceedings of ACL*. 3851–3863.
- [44] Yue Yu, Yinghao Li, Jiaming Shen, Hao Feng, Jimeng Sun, and Chao Zhang. 2020. Steam: Self-supervised taxonomy expansion with mini-paths. In *Proceedings of KDD*. 1026–1035.
- [45] Jieyu Zhang, Xiangchen Song, Ying Zeng, Jiaze Chen, Jiaming Shen, Yuning Mao, and Lei Li. 2021. Taxonomy completion via triplet matching network. In *Proceedings of AAAI*, Vol. 35. 4662–4670.
- [46] Yuchen Zhang, Amr Ahmed, Vanja Josifovski, and Alexander Smola. 2014. Taxonomy discovery for personalized recommendation. In *Proceedings of WSDM*. 243–252.

Appendix

A Implementation Details

TaxoBell is implemented using PyTorch with the baselines taken from the respective repositories of their original authors. All training and inference tasks were conducted on a 48GB A6000 GPU. We adopt bert-base-uncased as the default encoder. The projection head comprises two 2-layer MLPs (hidden size 64) with a dropout rate of 0.2. Optimization uses AdamW with learning rates 9×10^{-5} for BERT fine-tuning and 1×10^{-3} for the projection layers. We train with a batch size of 128 for up to 125 epochs. The number of hard negatives per query is 50 for SCIENCE, 50 for ENVIRONMENT, 10 for WORDNET, 20 for MESH, and 20 for FOOD. Unless noted, loss weights are 0.45 for $\mathcal{L}_{\text{asym}}$, 0.45 for $\mathcal{L}_{\text{sym}}$, and 0.10 for the volume regularizer. For all baseline comparisons, we set $\lambda = 0.3$ and $C=1.5$.

Table 9: Statistics of benchmark datasets. $|\mathcal{N}^0|$ and $|\mathcal{E}^0|$ denote the number of nodes and edges in the seed taxonomy, while $|D|$ is the taxonomy depth. For WordNet, values are averaged across 114 sub-taxonomies.

Dataset	$ \mathcal{N}^0 $	$ \mathcal{E}^0 $	$ D $
Single-parent taxonomies			
SemEval-Env	261	261	6
SemEval-Sci	429	452	8
WordNet	20.5	19.5	3
Multi-parent taxonomies			
SemEval-Food	1486	1533	8
MeSH	9710	10498	12

Table 10: Impact of the asymmetric optimization ($\mathcal{L}_{\text{asym}}$) on ENV, and WordNet benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Environment			WordNet		
	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O $\mathcal{L}_{\text{asym}}$	44.23	11.51	53.55	34.58	47.53	47.48
$\mathcal{L}_{\text{asym}}$	52.20	8.98	58.36	37.40	41.00	52.40
↑ %	+18.02	+21.98	+8.98	+8.16	+13.74	+10.36

B Benchmark Datasets

As discussed in Section 5.1.1, we evaluate TaxoBell on five public taxonomies, namely (i) Environment (ENV), (ii) Science (SCI), (iii) Food, (iv) WordNet, and (v) MeSH, summarized in Table 9. The dataset statistics are discussed in Table 9. ENV and SCI are small taxonomies that approximate hand-curated taxonomies by experts, WordNet is a medium-scale and domain-mixed taxonomy, reflecting noisier, application-agnostic settings in real-world, while Food and MeSH are large, multi-parent taxonomies that stress-test scalability and closely mirror real-world multi-parent settings where structures deviate from strict trees. Each dataset provides surface names and definitions for concepts. For evaluation, we build a seed taxonomy and hold out a set of query nodes by removing 20% of leaf nodes while ensuring each query’s gold parent remains in the seed. The remaining seed nodes supply self-supervision during training and serve as candidate anchors at inference.

C Baselines

We compare TaxoBell with a diverse set of the following baselines outlined in Section 5.1.2,

- **BERT+MLP** [17] encodes term surface forms with BERT and feeds the resulting vectors to a multi-layer perceptron to classify hypernym relations.
- **TaxoExpan** [33] represents an anchor node by encoding its ego network with a graph neural network and scores parent–child pairs through a log-bilinear feed-forward layer.
- **Arborist** [25] models heterogeneous edge semantics and trains with a large-margin ranking loss whose margin adapts dynamically.

- **BoxTaxo** [15] learns box embeddings and evaluates parent candidates with geometric and probabilistic losses derived from hyper-rectangle volumes.
- **TMN** [45] method employs subtasks, namely attaching the query to the parent and the child to the query, as auxiliary supervision signals for concept representation learning.
- **STEAM** [44] ensembles graph, contextual, and lexical-syntactic features in a multi-view co-training framework to score hypernymy links.
- **TaxoEnrich** [14] utilizes structural information through taxonomy-contextualized embeddings, enhancing position representations with a query-aware sibling aggregator.

TaxoEnrich and TMN are designed for taxonomy completion. They insert a query q between a parent–child pair and therefore score triplets $f(p, c, q)$. Our setting is taxonomy expansion, where q is attached as a leaf under a parent and no child c is available. For a fair comparison, we follow [40] and adapt these baselines by instantiating c with a dummy placeholder (e.g., a blank/sentinel token). This converts their triplet scorer to an effective leaf-attachment scheme while preserving their original scoring function.

D Evaluation Metrics

Given the query set C , let the predictions made by the model be $\{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_{|C|}\}$ and the ground-truth positions be $\{y_1, y_2, \dots, y_{|C|}\}$. Following Jiang et al. [14], Liu et al. [22], Manzoor et al. [25], Vedula et al. [36] and Yu et al. [44], we use the following metrics to evaluate the performance of the competing models.

- **Hit@k**: It measures the fraction of instances where the correct label appears among the top- k predictions. It quantifies how often the model successfully ranks the true class within its top- k guesses. Formally, a hit is counted if $y_i \in \hat{Y}_i^{(k)}$ for the i -th instance. Higher Hit@k indicates better retrieval or ranking performance.

$$\text{Hit@k} = \frac{1}{|C|} \sum_{i=1}^{|C|} \mathbb{I}(y_i \in \hat{Y}_i^{(k)}), \quad (14)$$

where $\hat{Y}_i^{(k)}$ denotes the set of top- k predicted labels for the i -th instance, and $\mathbb{I}(\cdot)$ is the indicator function that equals 1 if the ground-truth label y_i is present in the top- k predictions, and 0 otherwise.

- **Recall@k**: It measures the proportion of ground-truth labels that are correctly retrieved within the top- k predictions for each instance. It reflects the model’s ability to capture relevant labels among its top-ranked outputs. Formally, it is defined as:

$$\text{Recall@k} = \frac{1}{|C|} \sum_{i=1}^{|C|} \frac{|\{y \in \mathcal{Y}_i : y \in \hat{Y}_i^{(k)}\}|}{|\mathcal{Y}_i|}, \quad (15)$$

where $\mathcal{Y}_i$ is the set of true labels and $\hat{Y}_i^{(k)}$ the top- k predicted labels. Note that for single parent taxonomies, where each instance has exactly one ground-truth label ($|\mathcal{Y}_i| = 1$), Recall@k becomes equivalent to Hit@k.

- **Mean Rank (MR)**: It measures the average position (rank) of the correct label among all predicted scores. It evaluates how highly the model ranks the ground-truth label across all instances. A

Table 13: Effect of $\mathcal{L}_{\text{diverge}}$ on ENV, and WordNet benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Environment			WordNet		
	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O $\mathcal{L}_{\text{diverge}}$	44.23	10.40	59.27	35.32	61.23	39.12
$\mathcal{L}_{\text{diverge}}$	52.20	8.98	58.36	37.40	41.00	52.40
↑ %	+18.02	+13.65	−1.54	+5.89	+33.04	+33.95

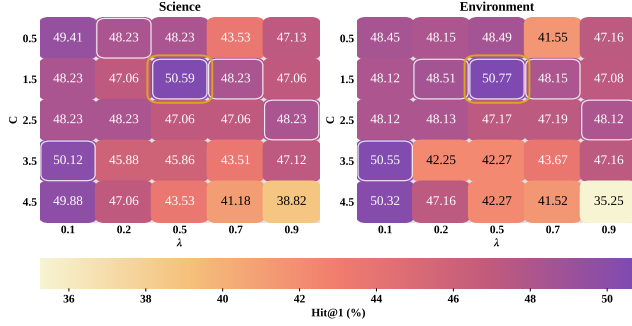


Figure 5: Heatmap of sensitivity of the $\mathcal{L}_{\text{diverge}}$ hyperparameters on Hit@1 for Science and Environment. We sweep the scale C (vertical axis) and the weight λ (horizontal axis).

Table 11: Impact of the symmetric optimization ($\mathcal{L}_{\text{sym}}$) on ENV, and WordNet benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Environment			WordNet		
	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O $\mathcal{L}_{\text{sym}}$	13.46	62.01	18.06	8.87	228.64	8.71
$\mathcal{L}_{\text{sym}}$	50.91	8.89	61.20	36.84	45.17	48.96
↑ %	+278.2	+85.7	+239.0	+315.3	+80.2	+462.0

Table 12: Effect of the volume regularization ($\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{clip}}$) on ENV, and WordNet benchmarks. “↓” indicates that lower values denote better performance. ‘W/O’ means *without*.

Method	Environment			WordNet		
	H@1	MR↓	MRR	H@1	MR↓	MRR
W/O	38.46	13.25	49.73	32.24	53.48	46.27
Reg	52.20	8.98	58.36	37.40	41.00	52.40
↑ %	+35.73	+32.23	+17.35	+16.00	+23.34	+13.25

lower mean rank indicates better performance, as correct labels appear earlier in the ranking. Formally, it is defined as:

$$\text{MR} = \frac{1}{|C|} \sum_{i=1}^{|C|} \text{rank}(y_i), \quad (16)$$

where $\text{rank}(y_i)$ denotes the position of the true label y_i in the sorted list of predicted scores for the i -th instance (rank 1 being the best).

- **Mean Reciprocal Rank (MRR):** It evaluates the average inverse rank of the correct label across all instances. It emphasizes not only whether the true label appears in the predictions but also how highly it is ranked. Higher MRR values indicate that correct labels are placed closer to the top of the ranked list. Formally, it is defined as:

$$\text{MRR} = \frac{1}{|C|} \sum_{i=1}^{|C|} \frac{1}{\text{rank}(y_i)}, \quad (17)$$

where $\text{rank}(y_i)$ denotes the position of the ground-truth label y_i in the model’s predicted ranking for the i -th instance.

- **Wu & Palmer (Wu&P) Similarity:** It measures the semantic similarity between two concepts based on their depth in a taxonomy and the depth of their lowest common ancestor (LCA). It quantifies the degree of relatedness between two nodes within a hierarchical structure. For single-parent taxonomies, where each node has a unique ancestor path, the measure is defined as:

$$\text{Wu\&P}(c_1, c_2) = \frac{2 \times \text{depth}(\text{LCA}(c_1, c_2))}{\text{depth}(c_1) + \text{depth}(c_2)}, \quad (18)$$

where $\text{depth}(c)$ denotes the distance of the concept c from the root, and $\text{LCA}(c_1, c_2)$ is their lowest common ancestor.

E Ablations

As a complement to Section 5.3, we report additional ablations on ENV and WordNet in Tables 10, 11, 12, and 13. The same pattern is evident in both benchmarks. Removing the asymmetric containment loss compromises hierarchical directionality, causing queries to drift to nearby but structurally incorrect anchors. Dropping the symmetric overlap loss weakens semantic cohesion and produces brittle neighborhoods, which is the most damaging setting and leads to large drops in ranking quality. Eliminating volume regularization allows covariances to collapse or explode, breaking the parent–child size relation and degrading all metrics. Adding the reverse-KL coverage term improves results relative to using alignment alone because it keeps parents broader than their children and stabilizes the training process. Taken together, these results demonstrate that overlap, directional containment, coverage control, and volume regularization work in tandem to produce calibrated Gaussian boxes and reliable attachments, and that removing any single component consistently degrades performance.

F Scaling and Dynamic Margin factor

To complement the ablations in Section 5.3, we tune the two hyperparameters in the divergence term $\mathcal{L}_{\text{diverge}}$, namely the scale C and the weight λ . The scale C controls how much broader a parent should be than its child and sets the required information gap. The weight λ sets the overall influence of $\mathcal{L}_{\text{diverge}}$ in the objective. Fig. 5 reports results for Science and Environment. A moderate C improves performance by enforcing a meaningful coverage gap. A very large C becomes too restrictive, degrading parent-child alignment. The weight λ shows a similar trade-off. Very small values are underregularized, while very large values push the model toward

Table 14: Qualitative examples of TaxoBell on the MeSH and Food benchmarks (Multi-Parent). For each benchmark, we show one correct and one incorrect attachment. Correct predictions are marked with ✓ while incorrect with ✗.

Query	Anchor (Gold)	Top Predictions
MeSH		
Copper	Transition elements, Heavy metals	Transition elements ✓ Tin ✗ Heavy metals ✓
Mitochondrial membrane transport proteins	Mitochondrial proteins, Membrane transport proteins	Cation transport proteins ✗ Electron transport chain-complex proteins ✗ NADH dehydrogenase ✗
Translational protein modification	Gene expression regulation, Protein biosynthesis	Peptide biosynthesis ✗ Translational peptide chain-termination ✗ Protein biosynthesis ✓
Food		
Frozen orange juice	Orange juice, Concentrate	Fruit juice ✗ Orange juice ✓ Grapefruit juice ✗
Mocha	Coffee, Flavorer	Dessert ✗ Charlotte ✗ Course ✗
Onion bread	Bread	Rye bread ✗ Bread ✓ Black bread ✗

a rigid geometric pattern that shrinks child variances and inflates parent variances. This reduces expressiveness and weakens semantic alignment. In practice, we find $C \approx 1.5$ and $\lambda \approx 0.5$ to be a good

operating point. These settings preserve the intuition that parents should be wider than children while keeping the embedding space well-calibrated.

G Case Study

As discussed in Section 5.5, we include additional multi-parent case studies for the MeSH and SemEval-Food benchmarks as shown in Table 14. On **MeSH**, *Copper* is correctly linked to both *Transition elements* and *Heavy metals*, indicating that the model can allocate probability mass to cover more than one valid parent, with Tin also ranked in top-3 because copper and tin are closely related metals that are often co-mentioned (e.g., bronze is an alloy of copper and tin), their definitions and usage strongly overlap, yielding a high symmetric-overlap score. In contrast, *Mitochondrial membrane transport proteins* is misattached to nearby transport categories because the query shares strong lexical and semantic cues such as *mitochondrial*, *membrane*, *transport*, and *proteins*, with several sibling classes whose MeSH definitions describe closely related processes, which boosts symmetric similarity across candidates and, under a shared “transport” ancestor, weakens the KL-based preference for the correct parent. On **SemEval-Food**, *Frozen orange juice* returns several reasonable parents (*Fruit juice*, *Orange juice*). This shows the model can capture both broad categories and specific products. It also ranks *Fruit juice* above the gold label *Concentrate*, so the top result looks *wrong* even though Fruit juice is arguably a better parent. In contrast, *Mocha* leans toward dessert-related parents, which likely comes from ambiguity between drinks and sweets. In general, the model does well on multi-parent cases when the shared meaning across parents is clear. Mistakes happen when sibling categories are very similar or the query is ambiguous. Better definitions, harder negatives from the same family, and tighter uncertainty control may further improve these cases.